PSOC™ Edge Trusted Firmware-M

Customer training workshop

January 2026

Pre-requisites

Training prerequisites

- Introduction to PSOC™ Edge and ModusToolbox™
  - For an introduction to PSOC™ MCUs, visit https://www.infineon.com/product-information/psocdeveloper
  - For PSOC™ Edge trainings, visit PSOC™ Edge E84 Training Collection

Development tools

- ModusToolbox™ v3.6
- Eclipse IDE for ModusToolbox™ v2025.4.0
- Edge Protect Security Suite v1.6
- ModusToolbox™ Programming Tools v1.5.0
- KIT_PSE84_EVAL_EPC2 BSP v1.0.0
- Terminal Emulator
- KIT_PSE84_EVAL

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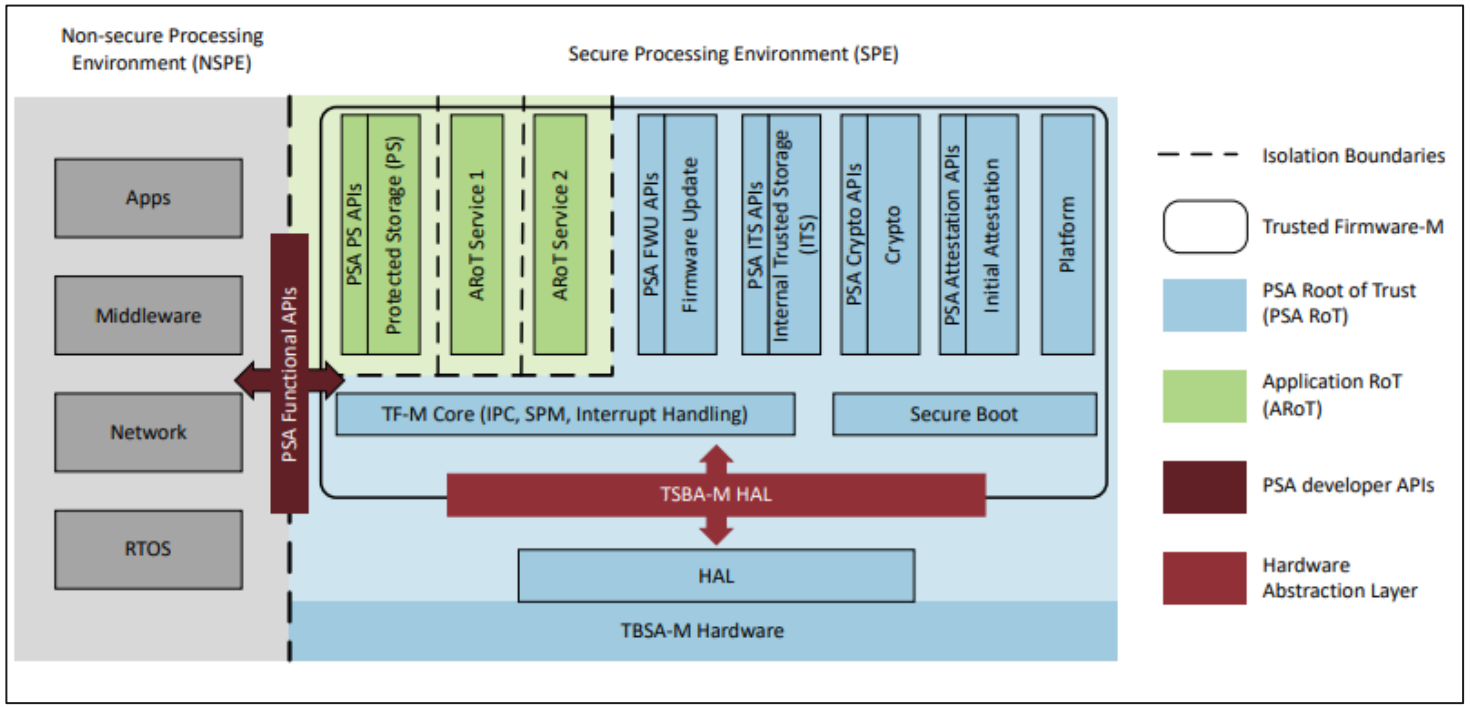
Trusted Firmware-M (TF-M) overview

Trusted Firmware-M (TF-M) provides a Secure Processing Environment (SPE) for Armv8-M, Armv8.1-M, and dual-core platforms. It serves as the reference implementation of the Platform Security Architecture (PSA), aligning with PSA Certified guidelines to enable devices to achieve PSA Certification.



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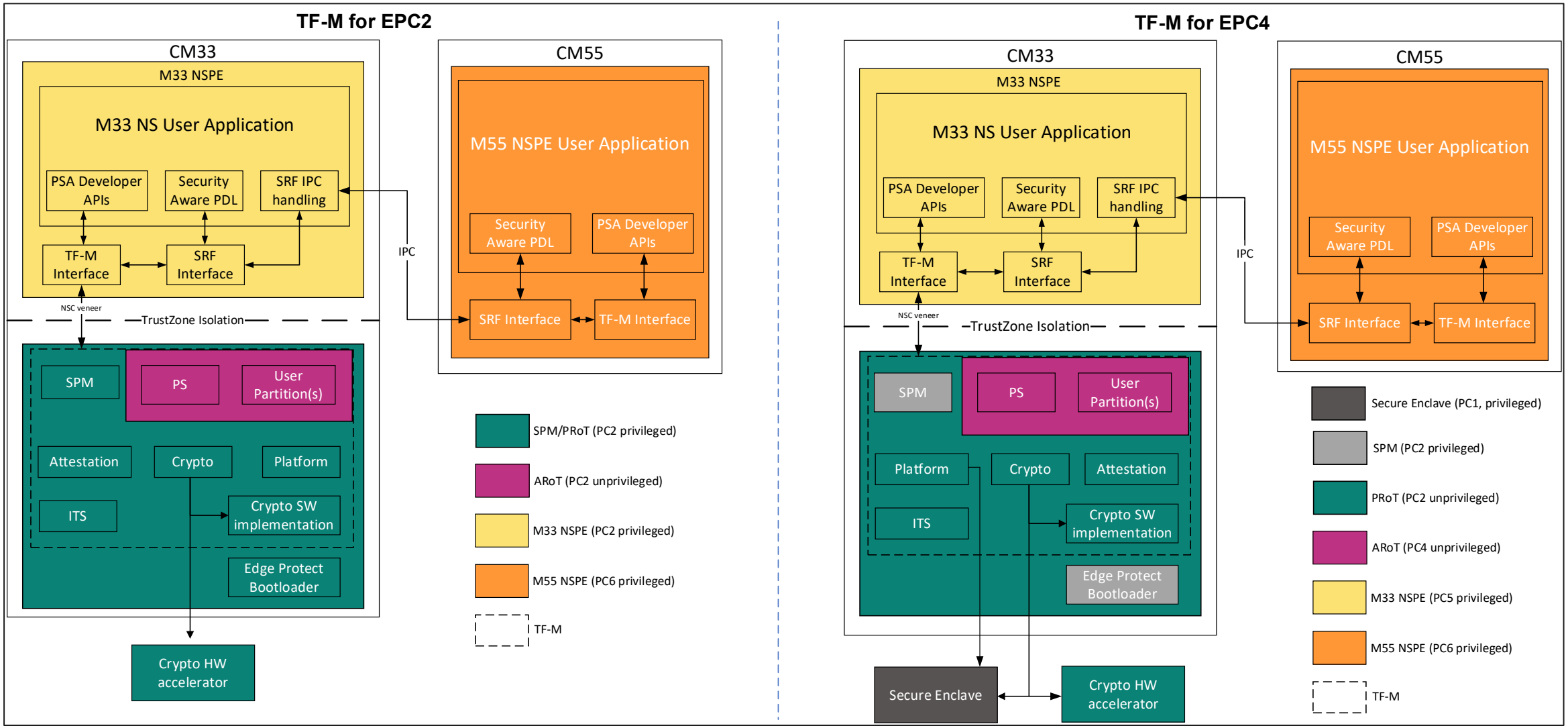
TF-M architecture



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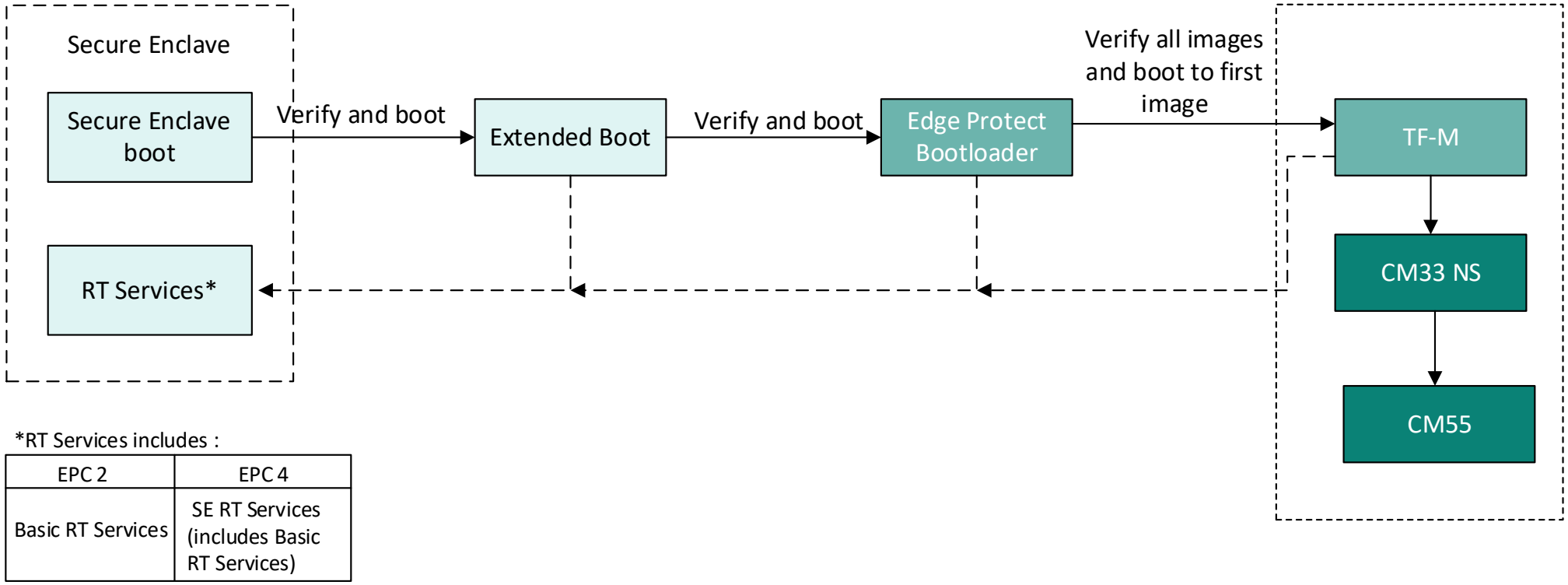
Boot flow with TF-M



*RT Services includes :

EPC 2	EPC 4
Basic RT Services	SE RT Services (includes Basic RT Services)

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Isolation levels

The PSA Firmware Framework-M (PSA FF-M) specification defines three isolation levels:

- Level 1 : Isolation boundary exists between SPE and NSPE
- Level 2 : Isolation boundary exists between SPE and NSPE, and between PSA RoT and App RoT
- Level 3 : Isolation boundary exists between SPE and NSPE, between PSA RoT and App RoT, and among individual App RoT partitions

Device compliance:

- EPC2: PSA L2 compliant and supports isolation level 2 (L2) and isolation level 3 (L3), with L2 being the default isolation level
- EPC4: Supports both L2 and L3, with L3 as default

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EPC 2 and EPC 4 privilege levels

Privilege level of EPC2 software component

Software component	NS attribute	PC	Priv attribute
Edge Protect Bootloader	S	PC2	Priv
TF-M core (SPM)	S	PC2	Priv
PSA RoT	S	PC2	Priv
App RoT	S	PC2	Unpriv
M33 NSPE	NS	PC2	Priv
M55 NS application	NS	PC6	Priv

Privilege level of EPC4 software component

Software component	NS attribute	PC	Priv attribute
SE RT services	S	PC1	Priv
Edge Protect Bootloader	S	PC2	Priv
TF-M core (SPM)	S	PC2	Priv
PSA RoT	S	PC2	Unpriv
App RoT	S	PC4	Unpriv
M33 NSPE	NS	PC5	Priv
M55 NS application	NS	PC6	Priv

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TF-M profiles

	Small (not supported)	**Medium**	**ARoT less (not supported)**	**Large**
Isolation level	Level 1	Level 2	Level 1	Level 3
Crypto capabilities	• Symmetric cipher suite • SHA 256 hashing	• Symmetric and asymmetric cipher suite • SHA 256 hashing	• Symmetric and asymmetric cipher suite • SHA 256 hashing	• Larger symmetric and asymmetric cipher suite • SHA 256/384/512 hashing
Services/partitions	• ITS • Initial Attestation (symmetric) • Crypto	• ITS • Initial Attestation (asymmetric) • PS • Crypto	• ITS • Initial Attestation (asymmetric) • FWU • Crypto	• ITS • Initial Attestation (asymmetric) • PS • Crypto • FWU (not supported)

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TF-M services: Cryptography

- TF-M offers a cryptography service implementing the PSA crypto API within a PSA RoT secure partition using the MbedTLS library

- The crypto service is accessible to applications in NSPE and other TF-M services in SPE via PSA Crypto APIs

- TF-M supports four build profiles, each with different functionality:
  - Small (not supported on PSOC™ Edge)
  - Medium
  - ARoT less (not supported on PSOC™ Edge)
  - Large

- By default, PSOC™ Edge EPC2 and EPC4 devices use the medium profile

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TF-M services: Cryptography – EPC 2

- The PSOC™ Edge EPC2 device implements cryptography using:
  - A crypto hardware accelerator for accelerated algorithms
  - Software implementation from the Mbed TLS library for other algorithms

- The choice between hardware and software implementation is determined at compile-time configuration

- By default, hardware accelerator support is enabled in TF-M for EPC2 devices. This support can be enabled or disabled by editing the application Makefile

- For key storage:
  - Persistent keys are stored in Internal Trusted Storage (ITS)
  - Volatile keys are stored in an SRAM region allocated to the Crypto partition

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TF-M services: Cryptography – EPC 4

The crypto implementation for the EPC4 device uses the following:

- Secured enclave (SE) for robust security
- Crypto hardware accelerator for accelerated algorithms (PSA L2 certifiable)
- Software implementation from the Mbed TLS library for other algorithms

Recommendation:

- Use Secure Enclave (SE) for crypto operations and keep the crypto hardware accelerator disabled. Enabling the crypto hardware accelerator prevents compliance with PSA L3/4

Key management:

- EPC4 uses the key lifetime attribute to determine whether SE or M33 software handles crypto operations. When SE is used, both persistent and volatile keys are stored inside SE
- For SHA256 hashing, two options are available with EPC4 device: SE or software implementation. Unlike other crypto operations, the SHA256 selection must be done at compile time. By default, SHA256 with SE is enabled

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TF-M services: Secured storage

Internal trusted storage (ITS)

- ITS is a PSA RoT service designed for storing the most security-critical device data, such as cryptographic keys, in the internal RRAM storage.
- In PSOC™ Edge, RRAM NVM is used to store the ITS assets. By default, ITS supports the storage of ten assets, with a maximum asset size of 512 bytes.

Protected storage (PS)

- PS is an application RoT service that offers encrypted storage of assets in the internal RRAM.
- PS uses AES-CCM based AEAD encryption to protect the assets.
- By default, PS supports the storage of ten assets, with a maximum asset size of 2 KB.

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TF-M services: Initial attestation

- The initial attestation service serves as a key mechanism to prove the genuineness of the device

- Upon request, TF-M can create an initial attestation token (IAT) that is compliant with the IETF entity attestation token (EAT) specification

- The attestation token contains a list of claims that serve as evidence to prove the identity of the device. The following list of claims are provided in the attestation token:
  - boot seed
  - challenge
  - client_idimplementation_id
  - instance_id
  - profile_id
  - security_lifecycle
  - sw_components (for each image) : measurement_description ; measurement_value ; signer_id ; sw_component_type ; sw_component_version

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TF-M services: Initial attestation – EPC 2/EPC 4

- The EPC2 device includes software component measurements for the following images:
  - CM33 SPE (TF-M)
  - CM33 NSPE
  - CM55 NSPE

- The EPC4 device includes software component measurements for the following images:
  - SE RRAM boot
  - SE RT services
  - Extended boot (L1 boot)
  - CM33 SPE (TF-M)
  - CM33 NSPE
  - CM55 NSPE

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TF-M platform services

TF-M for PSOC™ Edge offers the following platform services.

Non-volatile counters

TF-M offers up to three non-volatile counters of four bytes each to the NSPE.

System reset

The registers required to reset the system are protected and cannot be accessed by the NSPE. The system reset service provides a way to the NSPE application to reset the platform.

NSPE logging

The logging platform service can be used by the NSPE to log messages on the serial terminal using UART.

Security aware PDL

Infineon's port of TF-M has a Secure Request framework (SRF) integrated as part of the platform partition. It provides security aware driver implementation of SMIF, RTC, SysClk and SysPM modules.

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Default configurations for EPC 2/EPC 4

Configuration	EPC2	EPC4
Profile	Medium	Medium
Isolation level	Level 2	Level 3
Services/Partitions	• ITS • Initial attestation (asymmetric) • PS • Crypto • Platform	• ITS • Initial attestation (asymmetric with Secure Enclave) • PS • Crypto (with Secure Enclave) • Platform
NV counters location	RRAM	Secure Enclave

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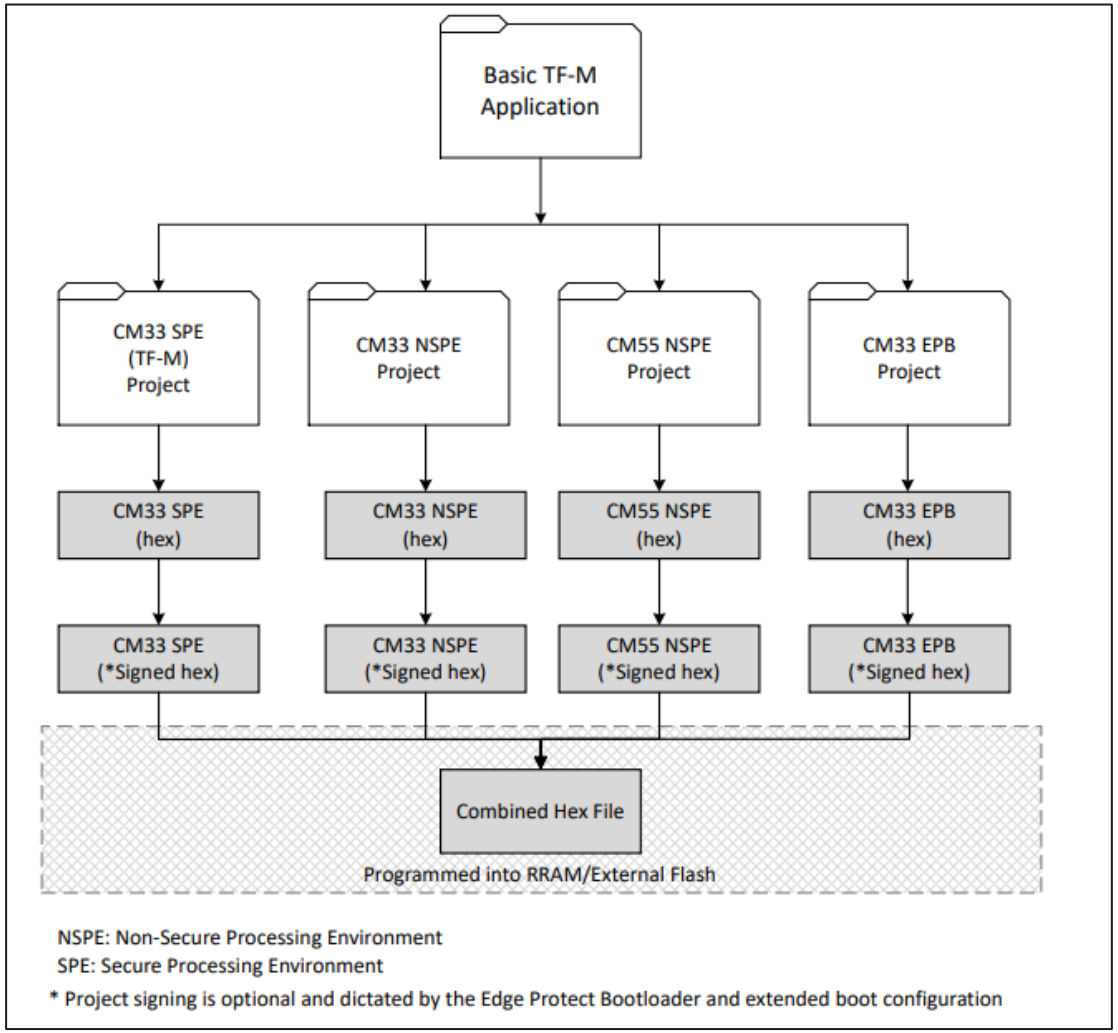
Application structure with TF-M



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Revision history

Document revision	Date	Description of change
**	2026-01-14	Initial release.

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